

## Slides 2.18: Buying Low, Rebalancing, and Return Math

Lesson 2.18, one day. Slides 1 to 12. Write “Round 8, 1932” and the prices on the board before class from page 2 of Handout 02.14 - The Gilded Age Market Kit.md and keep them covered until slide 3; the dispatch is on page 3.

### Slide 1: Do now

- In 1928, GAS was \$40 a share. Suppose it fell to \$20 and you bought 10 shares for \$200.
- If the price ever climbs back to \$40, how much do you make?
- Harder: what percent of your money did you make? Notes: 3 minutes. \$20 a share, \$200 on 10 shares. Percent: 200 divided by 200, which is 100 percent. You doubled your money.

### Slide 2: That is what buying low means

- It is also what everyone who sells at the bottom hands to the person who buys from them.
- Today: the one formula that tells you who won. Image: two figures in 1930s coats passing a stock certificate; the one handing it over looks scared, the one receiving it looks calm.

### Slide 3: Round 8. 1932. The bottom.

- RAIL 10. OIL 25. GAS 20. ELEC 10. STEEL 25. AUTO 8. RADIO 3.
- Prices have fallen for almost three years. On July 8, 1932, the Dow Jones average closed at 41, eighty nine percent below its 1929 peak.
- About one worker in four has no job. Thousands of banks have failed and the money in them is gone. (Our bank survives. Most did not.)
- General Motors sells for what a good shirt costs. Image: a 1932 bread line under a shop sign. Notes: Uncover the prices. Ledgers first: the Round 8 column and the holdings table. Then read the dispatch from kit page 3 and ask its question: everyone is selling and nobody is buying. If you have cash, what does that make this? 5 minutes. Do not open trading yet.

### Slide 4: The bottom, and buying low

- The bottom: the lowest price before a recovery. Nobody knows it until it is behind them.
- Buying low: buying a company after it fell, because you believe it will recover.
- The risk: some do not. The three tests still apply. A cheap price is not a reason. A dispatch with a number is. Image: a line graph with a V and a question mark at the bottom of the V.

### Slide 5: Rebalancing after a crash

- Your pie changed shape without you doing anything: the risky slices shrank.

- Rebalancing: buy a little of what fell, sell a little of what held, to get back to the pie you chose in Round 5.
- Housekeeping, not heroics. Image: a pie with two shrunken slices, an arrow, and the same pie restored. Notes: Students copy the three definitions to the journal notes box.

### Slide 6: The formula

- Percent return = (ending minus starting) divided by starting, times 100.
- The bottom of the fraction is always the starting value.
- A loss gets a minus sign. Image: the fraction drawn large: “ending minus starting” over “starting,” with “x 100” beside it and STARTING in bold.

### Slide 7: Worked example

- Started with \$1,000 in 1870. \$850 now.
- 850 minus 1,000 = minus 150.
- Minus 150 divided by 1,000 = minus 0.15.
- Times 100: minus 15 percent. Notes: Say “divided by the starting value” three times today.

### Slide 8: Return Math, Parts A and B

- Part A: four practice problems from the board’s prices, alone. Then the trick question: GAS, 40 to 20 and back.
- Part B: my own return, from my ledger’s Round 8 total, alone. And the company that hurt me most, 1928 price to 1932 price.
- Part C waits until after trading. Notes: 12 minutes for slides 4 to 8. The common error is dividing by the ending value. Watch for it at the desk.

### Slide 9: Trading, Round 8, then Part C

- Trading is open for four minutes at 1932 prices.
- If your pie changed shape in the crash, this is the round to fix it.
- If you are buying low, write which company and what you expect on Entry 8 first. If you are holding, write that too.
- At four minutes the market freezes. Nobody trades again until the closing bell.
- Then Part C with a partner: three investors from 1870 who never traded, at the peak, in the crash, and today. Notes: 10 minutes. Four for trading, six for Part C.

### Slide 10: Debrief Part C

- Safe, all in the bank: plus 19.5, plus 23.1, plus 26.8 percent. Never dropped.
- One company, all in RAIL: plus 40.0, minus 10.0, minus 80.0 percent.
- Mixed, four ways: plus 189.1, plus 117.9, plus 24.5 percent.
- Which would you want for college in ten years? For rent next month? Image: three lines on one graph across 1928, 1929 and 1932: a gentle rise, a cliff, a mountain that comes most of the way down. Notes: Say the honest thing: the bank won today by two points after 62 years, and the mixed ledger was ahead in every other round.

### Slide 11: Entry 8

- What I did (rebalance, buy low, or hold). Why. What happened.
- My percent return so far is \_\_\_\_ (from Part B).
- The company that hurt me most fell \_\_\_\_ percent from 1928 to 1932.
- My total in dollars at the freeze. Notes: 6 minutes.

### Slide 12: Closure

- Exit card: the percent return formula in words. My return today is \_\_\_\_ percent. The all-bank ledger's return today is \_\_\_\_ percent. One sentence: which is better, and better for what?
- Three investors read the last sentence.
- Next class: the closing bell, 1936. And what happened next to every company you own. Notes: 4 minutes. The market is frozen at Round 8. Do not say what 1936 prices look like.